Petty's equilateral-set conjecture in dimension five

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Abstract. Petty conjectured that every n-dimensional real normed space contains $n+1$ equidistant points. The conjecture was proved in dimensions three and four by Petty and Makeev, respectively. We prove the five-dimensional case: every five-dimensional real normed space contains six equidistant points. The proof uses a bordism of equilateral configurations and a homological obstruction.

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1. Introduction

A finite set S in a real normed space X is *equilateral* if there is a number $\lambda > 0$ such that

$$\|x - y\| = \lambda, \qquad x, y \in S, \quad x \neq y.$$

Let $e(X)$ denote the largest cardinality of an equilateral subset of X. Petty's equilateral-set conjecture [7] asserts that

$$e(X) \geq \dim X + 1$$

for every finite-dimensional real normed space. The Euclidean regular simplex shows that this proposed universal lower bound is sharp. Equilateral sets are antipodal [7]. The theorem of Danzer and Grünbaum [1] therefore yields $e(X) \leq 2^{\dim X}$. For background on equilateral sets and their relation to Minkowski geometry, see [9, 11].

The conjecture is immediate in dimension one. In the plane, choose a unit vector u and a continuous arc γ on the unit circle from u to $-u$. The function $t \mapsto \|\gamma(t) - u\|$ takes the values 0 and 2, and hence takes the value 1. This gives the equilateral triangle $\{0, u, \gamma(t)\}$. Petty [7] proved the three-dimensional case, and Kobos [4] gave an alternative proof. Makeev [6] proved the four-dimensional case. The present paper establishes the next case.

Theorem 1.1. *Every five-dimensional real normed space contains an equilateral set of six points. Equivalently, for every norm N on $\mathbb{R}^5$ there are $x_0, \dots, x_5 \in \mathbb{R}^5$ such that $x_0 = 0$ and*

$$N(x_i - x_j) = 1, \qquad 0 \leq i < j \leq 5.$$

The theorem concerns the existence of an equilateral set; it does not require a prescribed set to extend. This distinction is relevant because maximal equilateral sets can be small [7, 10]. Nor do we require the six points to be affinely independent. Configurations of affine rank four occur in the argument and must be retained.

We argue by contradiction. The absence of a six-point equilateral set persists under sufficiently small perturbations of a norm. A hypothetical counterexample can therefore be chosen smooth and strongly convex. Join it to the Euclidean norm and perturb the path, keeping the endpoints fixed. All centered, labeled unit-equilateral configurations along the path form a compact oriented eleven-manifold whose boundary is the Euclidean configuration space $O(5)$.

The main issue is the behavior of the configuration matrix at rank loss. The classification of Schürmann and Swanepoel [8, Theorem 2.3] excludes six equilateral points in a strictly convex three-dimensional normed space. Thus every matrix on the bordism has rank at least four. At rank four its kernel is a line of affine relations. Strict antipodality gives an explicit inequality for this line. Compactness places the relevant lines in a region with a uniform balance margin, for which we construct an equivariant graph model.

To finish, we cut the bordism at a small positive determinant level. Positive polar decomposition maps the resulting quotient bordism to $\mathrm{SO}(5)/C_5$, and is the identity on the Euclidean boundary. The other boundary factors through a bundle over a graph with fiber $\mathrm{SO}(5)$. Since $H_9(\mathrm{SO}(5);\mathbb{F}_5) = 0$, its tenth homology is generated by fiber classes. Each fiber maps to $\mathrm{SO}(5)/C_5$ with degree five, so this boundary has zero image over $\mathbb{F}_5$. It cannot cancel the nonzero fundamental class of the Euclidean boundary.

Section 2 recalls the transversality and homology tools. Section 3 carries out the smooth reduction, establishes the constraints on six-point configurations, and constructs the bordism and its relation-line graph. Section 4 computes the required homology, proves the determinant-cut obstruction, and concludes the proof of Theorem 1.1.

2. Preliminaries

All vector spaces are real and finite-dimensional. We fix auxiliary Euclidean inner products and, whenever determinants are used, orientations. The notation $|\cdot|$, orthogonal complements and adjoints refer to these inner products. A norm is *smooth* if it is C^∞ away from zero, *strictly convex* if its unit sphere contains no nontrivial line segment, and *strongly convex* if $D^2(N^2)(u)$ is positive definite for $u \neq 0$. Strong convexity implies strict convexity: on a segment avoiding zero, N^2 is strictly convex, and a segment through zero cannot lie in the unit sphere.

2.1. Regular fibers and transversality.

We recall the regular-value and transversality results used to construct the configuration bordism, together with the orientation convention needed to pass to its quotient.

For a smooth map $F : M \to W$ to a vector space, transversality to zero means that $DF_x : T_xM \to W$ is onto at every zero; the empty zero set is allowed [5, p. 143]. We use the following forms of the regular-value theorem [2, Chapter 1] and its boundary version [5, Problem 5-23].

Lemma 2.1 (Regular zero sets)**.** *Let M be a smooth m-manifold, possibly with boundary, and let W be an r-dimensional vector space. Suppose $F : M \to W$ is smooth and transverse to zero. If M has boundary, suppose also that $F|_{\partial M}$ is transverse to zero. Then $Z = F^{-1}(0)$ is a smooth submanifold of dimension $m - r$, possibly empty, with*

$$T_xZ = \ker DF_x, \qquad \partial Z = Z \cap \partial M.$$

The derivative gives the exact sequence of vector bundles

$$0 \longrightarrow TZ \longrightarrow TM|_Z \xrightarrow{DF} Z \times W \longrightarrow 0.$$

If M and W are oriented, the identification $\det TM|_Z \cong \det TZ \otimes \det W$ orients Z. If a finite group acts smoothly on M and linearly on W, preserves both orientations, and F is equivariant, then its action on Z preserves this orientation.

The orientation is the preimage orientation [5, Chapter 15]. The equivariant assertion follows by taking determinant lines in the exact sequence. Boundary orientations are taken with the outward normal first.

Sard's theorem [5, Theorem 6.10] supplies the regular values used below.

Lemma 2.2 (Sard's theorem)**.** *Let $F : M \to N$ be a smooth map between finite-dimensional smooth manifolds without boundary. The critical values of F have measure zero in every coordinate chart of N. Thus almost every $y \in N$ is a regular value, meaning that $DF_x : T_xM \to T_yN$ is onto for every $x \in F^{-1}(y)$. In particular, for a smooth real-valued function on M, every nonempty open interval contains a regular value.*

Lemma 2.3 (Parameter transversality)**.** *Let M be a smooth manifold without boundary, $P \subset \mathbb{R}^L$ open, W a finite-dimensional vector space, and $\Phi : M \times P \to W$ smooth. If Φ is transverse to zero, then $\Phi_a(x) = \Phi(x, a)$ is transverse to zero for almost every $a \in P$* [5, Theorem 6.35 and errata to p. 146].

Smooth cutoffs will be used to keep endpoint neighborhoods fixed and to localize perturbations. If K is compact in a smooth manifold and $K \subset U$ with U open, there is a smooth function χ with $0 \le \chi \le 1$, equal to one near K, whose support is a compact subset of U [5, Chapter 2]. A *bordism* between two closed smooth k-manifolds is a compact smooth $(k+1)$-manifold whose boundary is diffeomorphic to their disjoint union; here closed means compact without boundary.

2.2. **Homology of pairs, products and quotients.**

We collect the exact sequences and product formulas used in the homology calculation for bundles over the relation graph.

All homology groups have coefficients in $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$; we suppress these coefficients in this subsection. We write f_* for the homomorphism induced by f, and $\widetilde{H}_*$ for reduced homology.

For a nonempty space, $\widetilde{H}_j = H_j$ for $j > 0$ and $\widetilde{H}_0$ is the kernel of the augmentation $H_0 \to \mathbb{F}_5$. The next lemma collects the forms of homotopy invariance, exactness and excision that we use [3, Theorem 2.10 and Section 2.1], together with Mayer–Vietoris [3, Section 2.2, pp. 149–150].

Lemma 2.4. *Homology with coefficients in $\mathbb{F}_5$ has the following homotopy, exactness and excision properties.*

(i) *If $f, g : (X, A) \to (Y, B)$ are homotopic through maps of pairs, then $f_* = g_*$ on relative homology. Consequently a homotopy equivalence of pairs induces isomorphisms in all degrees. The same statements hold for absolute homology by taking $A = B = \varnothing$.*

(ii) *For every pair $A \subset X$, there is a natural long exact sequence*

$$\cdots \longrightarrow H_j(A) \xrightarrow{i_*} H_j(X) \xrightarrow{j_*} H_j(X, A) \xrightarrow{\partial} H_{j-1}(A) \longrightarrow \cdots . \tag{2.1}$$

Here i_* is induced by inclusion, j_* passes to relative classes, and ∂ takes a relative cycle to the class of its boundary in A. In particular, if $H_j(X, A) = 0$, then $H_j(A) \to H_j(X)$ is onto.

(iii) *If $Z \subset A \subset X$ and $\overline{Z} \subset \operatorname{int}_X A$, the inclusion of pairs induces an isomorphism*

$$H_j(X \setminus Z, A \setminus Z) \xrightarrow{\cong} H_j(X, A)$$

for every j.

(iv) *If $X = U \cup V$ with U, V open in X, there is a natural Mayer–Vietoris exact sequence*

$$\cdots \longrightarrow H_j(U \cap V) \xrightarrow{\alpha} H_j(U) \oplus H_j(V) \xrightarrow{\beta} H_j(X) \xrightarrow{\partial} H_{j-1}(U \cap V) \longrightarrow \cdots . \tag{2.2}$$

With i_U, i_V denoting inclusions from $U \cap V$ and j_U, j_V inclusions into X, the maps are

$$\alpha(a) = \big((i_U)_* a, -(i_V)_* a\big), \qquad \beta(b, c) = (j_U)_* b + (j_V)_* c.$$

A pair (X, A) is called *good* if A is a nonempty closed subspace that is a deformation retract of an open neighborhood in X. A CW pair is a CW complex together with a subcomplex. The quotient and wedge formulas below hold for good pairs [3, Section 2.1, pp. 124–126]; the product formulas are the field-coefficient and relative Künneth formulas [3, Section 3.B, p. 276].

Lemma 2.5. *The following quotient and product formulas hold for homology with coefficients in $\mathbb{F}_5$.*

(i) *For a good pair (X, A), the quotient map induces isomorphisms*

$$H_j(X, A) \cong \widetilde{H}_j(X/A), \qquad j \geq 0. \tag{2.3}$$

If finitely many based spaces (X_α, x_α) are good pairs, the inclusions into their wedge induce

$$\bigoplus_\alpha \widetilde{H}_j(X_\alpha) \cong \widetilde{H}_j\Big(\bigvee_\alpha X_\alpha\Big). \tag{2.4}$$

(ii) *For CW complexes X, Y, the cross product induces a natural isomorphism*

$$\bigoplus_{p+q=j} H_p(X) \otimes_{\mathbb{F}_5} H_q(Y) \xrightarrow{\cong} H_j(X \times Y). \tag{2.5}$$

For CW pairs (X, A) and (Y, B), its relative form is

$$\bigoplus_{p+q=j} H_p(X, A) \otimes_{\mathbb{F}_5} H_q(Y, B) \cong H_j(X \times Y, A \times Y \cup X \times B). \tag{2.6}$$

These formulas also hold for spaces and pairs homotopy equivalent to CW complexes and CW pairs, respectively.

(iii) *In particular, if X has CW homotopy type, $k \geq 1$ and $I = [0, 1]$, then*

$$H_j(X \times S^k) \cong H_j(X) \oplus H_{j-k}(X), \tag{2.7}$$

$$H_j(X \times I, X \times \partial I) \cong H_{j-1}(X), \qquad j \geq 1. \tag{2.8}$$

Groups of negative degree are understood to be zero.

The special cases ((iii)) and ((iii)) follow from ((ii)) and ((ii)). They use $H_0(S^k) = H_k(S^k) = \mathbb{F}_5$ and zero in all other degrees, together with $H_1(I, \partial I) = \mathbb{F}_5$ and zero in all other degrees [3, Section 2.1]. Compact manifolds have CW homotopy type [3, Corollary A.12], so these formulas apply to the compact Lie groups used below.

2.3. **Fundamental classes and finite coverings.**

We record how fundamental classes behave under boundaries and finite coverings. These identities provide the cancellation used in the proof of the main theorem.

We restore the coefficient notation in the formulas below. A connected closed oriented d-manifold has a fundamental class $[M]$, characterized by the specified local orientation at each point, and

$$H_d(M; \mathbb{F}_5) = \mathbb{F}_5[M], \qquad H_j(M; \mathbb{F}_5) = 0, \quad j > d$$

by [3, Theorem 3.26]. In particular, $[M] \neq 0$. For a compact oriented manifold Y with boundary, the relative fundamental class satisfies

$$\partial[Y, \partial Y] = [\partial Y]$$

under the connecting homomorphism, with the induced boundary orientation [3, Section 3.3, pp. 253–254]. For disconnected manifolds, these classes mean sums over their oriented components.

Lemma 2.6. *Let Y be a compact oriented $(d+1)$-manifold and $f : Y \to M$ continuous. Then*

$$\left(f|_{\partial Y}\right)_*[\partial Y] = 0 \quad \text{in } H_d(M; \mathbb{F}_5). \tag{2.9}$$

If $q : \widetilde{M} \to M$ is an orientation-preserving r-sheeted covering of connected closed oriented d-manifolds, then $q_*[\widetilde{M}] = r[M]$. For a finite regular covering of path-connected, locally path-connected spaces with deck group G, there is a transfer $\mathrm{tr} : H_j(M;\mathbb{F}_5) \to H_j(\widetilde{M};\mathbb{F}_5)$ satisfying

$$q_* \,\mathrm{tr} = |G|\,\mathrm{id}, \qquad \mathrm{tr}\, q_* = \sum_{g\in G} g_*. \tag{2.10}$$

If $|G|$ is invertible in $\mathbb{F}_5$ and every deck transformation acts trivially on homology, then q_* is an isomorphism with inverse $|G|^{-1}\,\mathrm{tr}$.

Proof. By ((ii)), the class $[\partial Y]$ vanishes in $H_d(Y;\mathbb{F}_5)$; applying f_* proves (2.6). An orientation-preserving r-sheeted covering has r positive local sheets, so its degree is r. The transfer [3, Section 3.G] sums all lifts of each singular simplex and commutes with the boundary operator. Projection gives $|G|$ copies of that simplex; in the other order, the lifts are its deck translates. This proves (2.6), and the last assertion follows by dividing both identities by $|G|$. □

Lemma 2.7 (Finite quotients)**.** *Let a finite group G act smoothly and freely on a smooth manifold Y, possibly disconnected and with boundary. Then Y/G has a smooth manifold structure for which the quotient map is a $|G|$-sheeted smooth covering, and*

$$\partial(Y/G) = (\partial Y)/G.$$

If Y is oriented and the action preserves orientation, the quotient has the unique orientation for which the covering is locally orientation preserving. The displayed boundary identification then preserves the induced orientations.

For connected manifolds without boundary this is the covering-space quotient theorem [5, Theorem 21.13]. In general, choose coordinate neighborhoods disjoint from their nontrivial translates, using half-space charts at boundary points. Their projections give the quotient charts and the asserted boundary identification; the orientations descend through the same charts.

3. Equilateral configurations and relation lines

We first reduce to smooth strongly convex norms and establish the geometric constraints on six-point configurations. These facts allow us to construct a bordism along a path of norms and a graph model for its relation lines, retaining the label action throughout.

3.1. Reduction to smooth strongly convex norms.

We show that a counterexample, if one exists, can be chosen smooth and strongly convex. We then establish the stability needed to perturb a path of such norms.

We give the space of norms the topology of uniform convergence on a fixed Euclidean unit sphere. Near each fixed norm N, this is equivalent to multiplicative convergence. To see this, write $x = \sum x_i e_i$; the triangle inequality gives $N(x) \le \sum |x_i| N(e_i) \le C|x|$, so N is continuous.

Its minimum a on the Euclidean unit sphere is positive. If $\sup_{|u|=1} |M(u) - N(u)| \le \varepsilon a$, homogeneity gives $(1-\varepsilon)N \le M \le (1+\varepsilon)N$. The converse follows from the boundedness of N on that sphere.

Lemma 3.1. *The norms on $\mathbb{R}^5$ admitting a six-point equilateral set form a closed set.*

Proof. Let $N_k \to N$, and translate and scale an N_k-equilateral set to obtain $x_0^k = 0$ and $N_k(x_i^k - x_j^k) = 1$ for $i \neq j$. There are constants $a, b > 0$ such that $a|v| \le N_k(v) \le b|v|$ for all sufficiently large k. Consequently $|x_i^k| \le a^{-1}$ and $|x_i^k - x_j^k| \ge b^{-1}$. Passing to a subsequence gives limits x_i with $N(x_i - x_j) = 1$. They are distinct and form the required equilateral set. □

Lemma 3.2. *Every norm on $\mathbb{R}^5$ is a limit of smooth strongly convex norms. Consequently, if Theorem 1.1 fails, it fails for such a norm.*

Proof. Fix a norm N and $0 < \eta < 1$. A norming functional at a unit vector u is a linear form f with $f(u) = 1$ and $|f(x)| \le N(x)$ for all x. Such a form exists by Hahn–Banach, applied to $f(tu) = t$ on $\mathbb{R}u$; see [11]. Each form remains greater than $1-\eta$ on a neighborhood of its contact point in the unit sphere. Compactness therefore supplies finitely many forms $f_1, \dots, f_L$ with

$$(1-\eta)N(x) \le \max_j |f_j(x)| \le N(x).$$

Append sufficiently small nonzero multiples of all coordinate forms and all forms $x_i + x_j$, keeping their dual norms at most one. Write L for the total number of forms after these additions. For an even integer $p \ge 4$ set

$$Q(x) = \sum_j f_j(x)^p, \qquad M(x) = Q(x)^{1/p}.$$

The linear map $x \mapsto (f_j(x))_j$ is injective, and M is the pullback of the ℓ_p norm under this map. Thus it is a norm; as $Q(x) > 0$ for $x \neq 0$, it is analytic away from zero. Moreover,

$$(1-\eta)N \le M \le L^{1/p} N.$$

Choose p so large that $L^{1/p} \le 1 + \eta$.

For $x \neq 0$,

$$D^2Q(x)[v,v] = p(p-1) \sum_j f_j(x)^{p-2} f_j(v)^2.$$

If this vanishes, choose a coordinate $x_i \neq 0$. The coordinate forms first give $v_j = 0$ whenever $x_j \neq 0$; the form $x_i + x_j$ then gives $v_j = 0$ for every remaining j. Thus $D^2Q(x)$ is positive definite. To deduce strong convexity of M, write $v = ax + w$ with $DM(x)[w] = 0$. Homogeneity gives

$$D^2(M^2)(x)[v,v] = 2a^2 M(x)^2 + 2M(x) D^2M(x)[w,w].$$

On $\ker DM(x)$, differentiation of $Q = M^p$ shows that $D^2Q = pM^{p-1}D^2M$. The displayed expression is therefore positive for $v \neq 0$. Letting $\eta \to 0$ proves the approximation claim. The final assertion follows from Lemma 3.1. □

For the construction below, approximation of a single norm is not enough. We will perturb an entire path of squared norms by functions which need not themselves be convex. The following observation explains why a sufficiently small perturbation remains a path of norms.

Lemma 3.3. *Let $Q : [0,1] \times (\mathbb{R}^5 \setminus \{0\}) \to (0,\infty)$ be smooth, even and homogeneous of degree two in the second variable. Assume $D_u^2 Q(t,u)$ is positive definite for every t and $u \neq 0$. Then $N^t = \sqrt{Q(t,\cdot)}$ is a smooth strongly convex norm. If $h_1, \dots, h_L$ are smooth even degree-two homogeneous functions away from zero, and χ is a smooth function of t, then the same conclusion holds for*

$$Q_{t,a}(u) = Q(t,u) + \chi(t) \sum_{j=1}^{L} a_j h_j(u)$$

whenever the vector $a \in \mathbb{R}^L$ is sufficiently small. Such a parameter ball can be chosen uniformly for all $t \in [0,1]$.

Proof. Extend each homogeneous function by zero at the origin. A smooth degree-two homogeneous function has gradient homogeneous of degree one. Bounds on the unit sphere show that this extension is C^1 at zero, with derivative zero. The restriction of $Q(t,\cdot)$ to any affine line has nonnegative second derivative away from a possible passage through zero, and its first derivative is continuous at that passage. It is therefore convex on the whole line. Thus $K_t = \{u : Q(t,u) \leq 1\}$ is convex and symmetric. Positivity and compactness of the sphere make K_t bounded with zero in its interior. Its Minkowski functional, defined by $p_{K_t}(u) = \inf\{r > 0 : u \in rK_t\}$, equals $\sqrt{Q(t,u)}$ by homogeneity. Convexity gives $p_{K_t}(u+v) \leq p_{K_t}(u) + p_{K_t}(v)$: divide $u+v$ by the sum of the two gauges and use a convex combination of points in K_t. Symmetry and boundedness give the other norm axioms. Its strong convexity is the assumed Hessian condition.

On the compact set $[0,1] \times S^4$, choose $c, \lambda > 0$ with

$$Q(t,u) \geq c, \qquad D_u^2 Q(t,u)[v,v] \geq \lambda |v|^2.$$

The values and Hessians of the finitely many $\chi(t)h_j$ are bounded there. For $|a|$ sufficiently small their total contributions have absolute value at most $c/2$ and Hessian operator norm at most $\lambda/2$. This preserves positivity and positive-definite Hessians on the time-sphere product, and homogeneity extends both properties to all nonzero u. The first part then applies to every $Q_{t,a}$. □

3.2. **Configuration matrices and affine relations.**

We represent centered six-point configurations by linear maps and derive the rank and relation inequalities imposed by strict convexity. These restrictions will control the singular matrices in the bordism.

Write $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^6$ and set

$$U = \Big\{ c \in \mathbb{R}^6 : \sum_i c_i = 0 \Big\}, \qquad b_i = \frac{e_i - \frac{1}{6}\mathbf{1}}{\sqrt{2}}, \quad 1 \leq i \leq 6. \tag{3.1}$$

Direct calculation gives $\langle b_i, b_j \rangle = \frac{1}{2}(\delta_{ij} - \frac{1}{6})$, so the b_i form a centered regular simplex of side length one. Also $\sum_i a_i b_i = 0$ holds exactly when all a_i are equal. Thus they span U, with sole relation $\sum_i b_i = 0$. Hence each centered tuple $y = (y_1, \ldots, y_6)$ in $\mathbb{R}^5$ has a unique matrix $A_y \in \mathrm{Hom}(U, \mathbb{R}^5)$ satisfying

$$A_y b_i = y_i. \tag{3.2}$$

This is a linear identification of the centered configuration space with $\mathrm{Hom}(U, \mathbb{R}^5)$. Since the centroid is zero, the affine span of the tuple is the linear span of its vertices, namely $\operatorname{im} A_y$. Consequently $\operatorname{rank} A_y$ equals its affine dimension, and

$$\ker A_y = \Big\{ c \in \mathbb{R}^6 : \sum_i c_i = 0, \quad \sum_i c_i y_i = 0 \Big\},$$

since $\sum_i c_i b_i = c/\sqrt{2}$ for $c \in U$.

We fix an oriented isometry $U \cong \mathbb{R}^5$ and identify the space of orientation-preserving isometries $U \to \mathbb{R}^5$ with $\mathrm{SO}(5)$.

The group S_6 acts orthogonally on U by coordinate permutations. Write ρ for this representation, and use the same notation for its restriction to $C_5 = \langle (12345) \rangle$, which fixes label six. For $\sigma \in S_6$, the relabeling action $(\sigma \cdot y)_i = y_{\sigma^{-1}(i)}$ satisfies $A_{\sigma \cdot y} = A_y \rho(\sigma)^{-1}$, since $\rho(\sigma) b_i = b_{\sigma(i)}$.

For $c \in U$ define

$$S(c) = \frac{1}{2} \sum_i |c_i|, \qquad \mathrm{osc}(c) = \max_i c_i - \min_i c_i.$$

For a nonzero zero-sum vector, $S(c)$ is both its total positive mass and its total negative mass in absolute value.

Lemma 3.4. *Let $y_1, \ldots, y_6$ be equilateral in a strictly convex norm. They are strictly antipodal: for every $i \neq j$ there is a linear functional uniquely maximal at y_i and uniquely minimal at y_j. Every nonzero affine relation c among these points satisfies*

$$\mathrm{osc}(c) < S(c). \tag{3.3}$$

Moreover, distinct unordered edges have distinct directions modulo sign.

Proof. Normalize all distances to one and choose a norming functional ℓ of $y_i - y_j$. Strict convexity implies that $y_i - y_j$ is the only unit vector where ℓ takes the value one. For $k \notin \{i, j\}$, the support inequality gives $\ell(y_k - y_j) < 1 = \ell(y_i - y_j)$ and $\ell(y_i - y_k) < 1 = \ell(y_i - y_j)$, since the corresponding unit vectors are distinct. Rearranging gives

$$\ell(y_j) < \ell(y_k) < \ell(y_i).$$

Thus the tuple is strictly antipodal. In particular, the difference between the maximum and minimum of ℓ on the vertices is one.

For a nonzero affine relation c, choose i with $c_i = \max_k c_k > 0$ and j with $c_j = \min_k c_k < 0$. Normalize the corresponding affine functional to take values $u_i = 1$, $u_j = 0$, and $0 < u_k < 1$ at

every other vertex. The relation gives $\sum_k c_k u_k = 0$. Therefore

$$c_i \le \sum_{c_k>0} c_k u_k = \sum_{c_k<0} (-c_k) u_k \le S(c) + c_j.$$

At least one inequality is strict: otherwise i and j are the only nonzero coefficients, which would imply $y_i = y_j$. This proves (3.4).

Finally, parallel unit edges agree after orienting them suitably. A norming functional for their common vector is uniquely maximal and minimal at the endpoints of each edge, as just proved. The ordered endpoints must therefore agree. □

Lemma 3.5. *Every six-point equilateral configuration in a strictly convex norm on $\mathbb{R}^5$ has affine rank at least four. At rank four, its unique relation line satisfies (3.4).*

Proof. By the classification theorem of Schürmann and Swanepoel [8, Theorem 2.3], a normed three-space with six equilateral points has either the one-skeleton of an affine regular octahedron on its unit sphere, or a two-dimensional face of its unit ball. Each alternative is incompatible with strict convexity. If the affine rank of our tuple were at most three, translating its affine span into a three-dimensional subspace and restricting the norm would give a contradiction. The restricted unit sphere is the intersection of the original unit sphere with that subspace, so it too contains no nontrivial segment. The last assertion follows from Lemma 3.4. □

3.3. **The configuration bordism.**

We use the preceding perturbation and geometric lemmas to construct a compact oriented bordism from the complete zero set of the distance equations. We also determine its label action and its Euclidean fiber.

Let

$$V = \Big\{(y_1, \dots, y_6) \in (\mathbb{R}^5)^6 : \sum_i y_i = 0\Big\}, \qquad W = \mathbb{R}^{15},$$

where the coordinates of W are indexed by unordered pairs of labels. On the open subset V° of tuples with distinct entries define

$$F_N(y) = \big(N(y_i - y_j)^2 - 1\big)_{i<j}. \tag{3.4}$$

A smooth norm is called *regular* here if $D_y F_N$ is onto at every zero. The empty zero set is regular by convention. The symmetric group S_6 acts by $(\sigma \cdot y)_i = y_{\sigma^{-1}(i)}$ and permutes the edge coordinates of W. These actions make F_N equivariant. The next lemma applies parameter transversality to the complete zero set of (3.3).

Lemma 3.6. *Let N^0, N^1 be regular smooth strongly convex norms on $\mathbb{R}^5$. There is a smooth path N^t, constant near its endpoints, for which*

$$\mathcal{B} = \{(t, y) \in [0,1] \times V^\circ : F_{N^t}(y) = 0\}$$

is a compact oriented smooth eleven-manifold, with boundary the two endpoint fibers. The label action of S_6 is free. Its subgroup $C_5 = \langle (12345) \rangle$ preserves the orientation. The Euclidean norm is regular, and its zero fiber is equivariantly diffeomorphic to $O(U, \mathbb{R}^5)$.

Proof. We construct a finite-dimensional family of norm perturbations, apply Lemma 2.3, and then identify its boundary and orientation.

Step 1. Construct a path with fixed endpoints and uniform bounds.

Choose a smooth $\eta : [0,1] \to [0,1]$ equal to zero on $[0,1/3]$ and one on $[2/3,1]$, and set

$$Q_t = (1-\eta(t))(N^0)^2 + \eta(t)(N^1)^2.$$

By Lemma 3.3, $\sqrt{Q_t}$ is a smooth strongly convex norm. On the compact time-sphere product, uniform equivalence to the Euclidean norm gives constants $a_0, b_0 > 0$ with $a_0|u| \le \sqrt{Q_t(u)} \le b_0|u|$. The same comparison, with a_0 decreased and b_0 increased once, holds for every sufficiently small perturbation on the sphere. At a unit-equilateral tuple,

$$b_0^{-1} \le |y_i - y_j| \le a_0^{-1}, \qquad |y_i| = \Big|\frac{1}{6}\sum_j (y_i - y_j)\Big| \le \frac{5}{6a_0}. \tag{3.5}$$

Consequently all zeros of all sufficiently small perturbations lie in one fixed compact subset $D \subset [0,1] \times V^\circ$. This excludes both escape to infinity and collisions. In particular the unperturbed zero set is compact.

Step 2. Perturb the fifteen edge equations independently.

Choose a smooth χ equal to one on $[1/3,2/3]$ and zero on $[0,1/4]\cup[3/4,1]$. Fix an unperturbed zero (t,y) with $t \in [1/3,2/3]$. For each unordered edge $e = \{i,j\}$ write $v_e = y_i - y_j$ with either choice of sign. By Lemma 3.4 the fifteen points $[v_e] \in \mathbb{RP}^4$ are distinct. Choose smooth even functions θ_e on S^4, each equal to one at $\pm v_e/|v_e|$, whose supports are pairwise disjoint. Smooth bump functions exist in these neighborhoods [5, Chapter 2]; averaging each function with its antipodal pullback makes it even. Define

$$h_e(u) = |u|^2 \theta_e(u/|u|), \qquad u \neq 0.$$

For the perturbation $Q_t + \chi(t)\sum_e a_e h_e$, differentiation of the residual on edge e' with respect to a_e gives

$$\chi(t) h_e(v_{e'}) = |v_{e'}|^2 \delta_{ee'}.$$

Thus the parameter derivative is onto W at this zero, and remains onto nearby. Compactness of the zero set in the middle interval gives finitely many such collections. List all their functions as $h_1, \dots, h_L$ and set

$$Q_{t,a}(u) = Q_t(u) + \chi(t)\sum_{j=1}^{L} a_j h_j(u), \tag{3.6}$$

$$\Phi(t,y,a) = \big(Q_{t,a}(y_i - y_j) - 1\big)_{i<j}.$$

The parameters belong to a small open ball $P \subset \mathbb{R}^L$. For $a = 0$ the total derivative of Φ is onto at every zero in the middle interval. Outside that interval, Q_t is an endpoint squared norm and its configuration derivative is onto by endpoint regularity. Hence $D\Phi$ is onto at every zero with $a = 0$.

Step 3. Choose a parameter giving the required smooth bordism.

Lemma 3.3 lets us shrink P so that every $N_{t,a} = \sqrt{Q_{t,a}}$ is a strongly convex norm, with the uniform bounds used in (3.3). Shrink P again so that Φ is transverse to zero throughout $[0,1]\times V^\circ \times P$. To justify this last choice, suppose that nontransverse zeros existed for a sequence $a_k \to 0$. Their (t,y) coordinates lie in D, so a subsequence converges to (t_*, y_*). This is a zero at $a = 0$, where $D\Phi$ is onto by Step 2. Surjectivity is open, a contradiction. This argument accounts for all possible new zeros of the perturbed path, not only those in one local configuration chart.

Apply Lemma 2.3 with $M = (0,1) \times V^\circ$ and the family (3.3). Choose a regular parameter $a \in P$ and fix $N^t = N_{t,a}$. By Lemma 2.1, the full interior zero set is a smooth manifold of dimension $1 + \dim V - \dim W = 1 + 25 - 15 = 11$. On the endpoint collars $\chi = 0$ and Q_t is constant. The zero set there is exactly an interval times the regular endpoint fiber. Therefore the complete zero set in $[0,1]\times V^\circ$ is a smooth manifold with boundary exactly the two endpoint fibers. It is compact by (3.3).

Step 4. Prove freeness and preservation of orientation.

For any norm, relabeling a tuple simply permutes the edge residuals. Thus every perturbation above is S_6-equivariant, even though no spatial rotational invariance is imposed on the bump functions. No nonidentity label permutation fixes a tuple of distinct points, so the S_6-action on $\mathcal{B}$ is free.

Orient $\mathbb{R} \oplus V$ and W. Along the regular zero set the derivative gives the equivariant exact sequence

$$0 \longrightarrow T\mathcal{B} \longrightarrow \mathcal{B} \times (\mathbb{R} \oplus V) \longrightarrow \mathcal{B} \times W \longrightarrow 0. \tag{3.7}$$

Writing D for this surjective derivative, the Euclidean metrics give the smooth equivariant right inverse $D^*(DD^*)^{-1}$. Thus the sequence splits and its determinant line defines an orientation of $\mathcal{B}$. The generator $\sigma = (12345)$ has determinant one on U: its determinant on $\mathbb{R}^6$ is one and it fixes $\mathbf{1}$. Hence its determinant on $V \cong \operatorname{Hom}(U, \mathbb{R}^5)$ is $(\det \rho(\sigma))^5 = 1$. On W, the fifteen edges divide into three orbits of length five: the five edges incident to label six, the adjacent pairs in the five-cycle, and its nonadjacent pairs. The determinant is again one. The time action is trivial. Taking determinant lines in (3.3) shows that the action on $T\mathcal{B}$ preserves orientation. The boundary collars also give the usual induced orientations on the endpoint fibers.

Step 5. Identify the Euclidean fiber and prove regularity.

For a centered Euclidean unit-equilateral tuple set $r_i = |y_i|^2$ and $R = \sum_i r_i$. Summing the five distances from y_i to the other vertices gives $6r_i + R = 5$. All r_i are therefore equal, $r_i = 5/12$, and $\langle y_i, y_j \rangle = -1/12$ for $i \neq j$. This is precisely the Gram matrix of the b_i in (3.2). The linear map $A_y b_i = y_i$ is consequently an isometry. Conversely each isometry gives such a tuple, and the linear dictionary (3.2) makes this identification an equivariant diffeomorphism.

To check regularity, suppose weights $\lambda_{ij} = \lambda_{ji}$ annihilate the fifteen edge differentials on V. Put $g_i = \sum_{j\neq i} \lambda_{ij}(y_i - y_j)$. The annihilation condition is $\sum_i \langle g_i, z_i \rangle = 0$ for every centered tangent tuple (z_i). Taking $z_i = v$, $z_j = -v$ shows that all g_i are equal. Their sum is zero, so

each $g_i = 0$. At a fixed simplex vertex the five incident edges form a basis of $\mathbb{R}^5$, and therefore every incident weight vanishes. Thus all $\lambda_{ij} = 0$, proving that the edge differential is onto. □

The construction uses every zero of the distance equations. In particular, configurations are retained when their matrices lose rank. By Lemma 3.5, the only possible ranks are four and five.

3.4. **A graph model for the relation-line space.**

We construct an equivariant graph model for compact regions of the admissible relation-line space. This reduces the later homology calculation to bundles over a finite graph.

Normalize a nonzero relation by $S(c) = 1$. Define

$$X_{\mathrm{bal}} = \{c \in U : S(c) = 1,\ \mathrm{osc}(c) < 1\}, \qquad K_{\mathrm{bal}} = X_{\mathrm{bal}}/(c \sim -c). \tag{3.8}$$

The second space parametrizes lines satisfying (3.4). For $0 < \delta < 1/6$ put

$$X_\delta = \{c \in U : S(c) = 1,\ \mathrm{osc}(c) \le 1 - \delta\}, \qquad K_\delta = X_\delta/(c \sim -c).$$

These compact spaces exhaust K_{bal} as $\delta \downarrow 0$. Here K_{bal} is viewed as the open subset of $\mathbb{RP}(U)$ defined by $\mathrm{osc}(c)/S(c) < 1$. Normalization $c \mapsto c/S(c)$ identifies its double cover with X_{bal}. In particular the topologies in (3.4) are the usual projective and quotient topologies. Every compact subset of K_{bal} lies in some K_δ, since its positive continuous balance margin has a positive minimum. Let Γ have one vertex for each subset $I \subset \{1, \dots, 6\}$ of size 2, 3 or 4, and an edge between I and J when $I \subset J$ and $|J| = |I| + 1$. Complementation acts on Γ; write $\bar{\Gamma} = \Gamma/(I \sim I^c)$.

Lemma 3.7. *For every $0 < \delta < 1/6$ there is an $S_6 \times \{\pm 1\}$-equivariant homotopy equivalence $X_\delta \simeq \Gamma$, where -1 acts by complementation on Γ. Consequently $K_\delta \simeq \bar{\Gamma}$ C_5-equivariantly. The C_5-actions on K_{bal} and $\bar{\Gamma}$ are free. The connected graph $\bar{\Gamma}/C_5$ has five vertices and twelve edges.*

Proof. We divide the proof into three steps.

Step 1. Determine the sign-cell intersections.

If $c \in X_\delta$ has p positive and q negative coordinates, then

$$\mathrm{osc}(c) \ge \frac{1}{p} + \frac{1}{q}.$$

If $p = 1$ or $q = 1$, the right side exceeds one. If $p + q \le 4$, then $1/p + 1/q \ge 4/(p+q) \ge 1$. Both possibilities contradict $\mathrm{osc}(c) < 1$. Hence $p, q \ge 2$ and $p + q \ge 5$, so at most one coordinate vanishes. For each allowed sign set I define the compact convex polytope

$$C_I = \left\{ c \in U : \begin{array}{c} \displaystyle\sum_{i \in I} c_i = 1, \quad c_i \ge 0,\ i \in I, \quad c_j \le 0,\ j \notin I, \\ c_i - c_j \le 1 - \delta, \quad i \in I,\ j \notin I \end{array} \right\}.$$

The defining conditions are linear equalities and inequalities, and $|c_i| \le 1$, so C_I is indeed compact and convex. These polytopes cover X_δ, by assigning a possible zero coordinate to either sign. If C_I and C_J meet, all coordinates in $I \triangle J$ vanish. Hence distinct cells meet only

for an edge of Γ, and no three meet. Conversely, if $I \subset J$ and $|J| = |I| + 1$, their intersection contains the point with coordinate zero on $J \setminus I$, value $1/|I|$ on I, and value $-1/(5-|I|)$ outside J. Its oscillation is $5/6 < 1 - \delta$.

Step 2. Construct equivariant maps in both directions.

We give an explicit equivariant homotopy equivalence. Map the vertex I to the point b_I with coordinates $1/|I|$ on I and $-1/(6 - |I|)$ on I^c. Map the midpoint of the edge $I - J$ to the point b_{IJ} just described, and map each half-edge linearly. The resulting map $f : \Gamma \to X_\delta$ is equivariant. Its half-edge images lie in the corresponding convex cells. Indeed the oscillation of b_I is $3/4$ for $|I| = 2, 4$ and $2/3$ for $|I| = 3$, while that of b_{IJ} is $5/6$; all are at most $1 - \delta$. Convexity keeps every joining segment in the appropriate cell.

For an edge $e = I - J$ write $E_e = C_I \cap C_J$. The sets E_e are pairwise disjoint and compact. Choose $\eta > 0$ so small that their η-neighborhoods are disjoint, miss all nonincident cells and all points b_I, and, for each e, miss every half-edge image other than the two belonging to e. Such a choice is possible: a half-edge image meets the shared faces only at its own midpoint, as its other coordinates retain their strict signs. The distance and the finite collection of conditions are invariant under permutations and sign reversal.

For $c \in C_I$ outside these neighborhoods put $g(c) = I$. If c is within distance η of E_e for an incident edge, put $g(c)$ on that half-edge at fraction $1 - \operatorname{dist}(c, E_e)/\eta$ of the way from I to the midpoint. There is at most one such edge. On E_e the two definitions agree at the midpoint. This defines a continuous equivariant map $g : X_\delta \to \Gamma$. More explicitly, its restriction to each C_I is continuous: at distance η the half-edge parameter tends to zero, and on a shared face it tends to the midpoint from either cell. Since the cells form a finite closed cover and the cellwise maps agree on their intersections, they paste to a continuous map on X_δ.

Step 3. Prove the homotopy equivalence and freeness.

Both c and $f(g(c))$ lie in the same cell, and in the same shared face when c lies on one. The formula $H(c, s) = (1 - s)c + sf(g(c))$ is therefore a well-defined homotopy in X_δ. This point matters because X_δ itself need not be convex: interpolation is performed within a single cell, where the normalization and sign conditions remain valid. On each half-edge, the chosen neighborhood separations ensure that gf stays in that half-edge and fixes its endpoint and midpoint. Linear interpolation in this interval gives $gf \simeq \mathrm{id}$. The homotopies agree at vertices and midpoints and commute with permutations and sign reversal.

Complementation fixes neither a vertex nor an edge of Γ. Thus the equivalence descends to $K_\delta \simeq \bar{\Gamma}$. There are $15+20+15 = 50$ vertices and $15 \cdot 4 + 20 \cdot 3 = 120$ edges before complementing, so $\bar{\Gamma}$ has 25 vertices and 60 edges. The graph is connected: two-element subsets can be joined by exchanging one element through a three-element superset, and the remaining vertices join to these.

No nonidentity element of C_5 fixes a vertex of $\bar{\Gamma}$. An invariant subset is a union of the five-element orbit and the singleton orbit, so its size is 0, 1, 5 or 6. If an element carried I to I^c, its square would fix I, giving the same contradiction. An odd-order action cannot interchange the endpoints of a fixed edge. The action on the graph is therefore free. Likewise, a fixed real

projective line for a nontrivial five-cycle has a representative c with $\rho(\sigma)^r c = \alpha c$. Since α is real and $\alpha^5 = 1$, it equals one. The first five coordinates are equal and the zero-sum condition forces the sixth to be minus five times their value. Thus the representative is proportional to $(1, 1, 1, 1, 1, -5)$. This vector has $\operatorname{osc}(c)/S(c) = 6/5$, so the line is outside K_{bal}. Dividing the graph counts by five finishes the proof. □

4. **Proof of the main results**

We prove Theorem 1.1 by combining the configuration bordism with the graph model from Section 3. First we show that the boundary associated with the relation graph contributes zero to tenth homology. We then cut at a regular determinant level and compare this contribution with the nonzero Euclidean boundary class.

4.1. **Homology of bundles over the relation graph.**

We compute the homology groups needed for the graph model and show that its associated bundle maps trivially to the quotient isometry group in degree ten.

The group $\mathrm{SO}(5)$ is a compact connected Lie group of dimension ten [3, Section 3.D].

Lemma 4.1. *One has $H_9(\mathrm{SO}(5); \mathbb{F}_5) = 0$. If $E \to T$ is a fiber bundle over a finite graph with fiber $\mathrm{SO}(5)$, then $H_{10}(E; \mathbb{F}_5)$ is generated by the images of the fundamental classes of fibers over the vertices of T.*

Proof. The standard quaternionic covering [3, Section 3.D, p. 294] is

$$S^3 \times S^3 \longrightarrow \mathrm{SO}(4), \qquad (a, b) \longmapsto (x \longmapsto axb^{-1}).$$

Its kernel is $\{(1, 1), (-1, -1)\}$. Indeed, if $axb^{-1} = x$ for all quaternions x, then $a = b$ by taking $x = 1$, and a commutes with all quaternions, so $a = b = \pm 1$. At the identity its differential sends a pair of imaginary quaternions (α, β) to the skew-adjoint map $x \mapsto \alpha x - x\beta$. If this map is zero, then $\alpha = \beta$ is both imaginary and central, so it is zero. The differential is therefore an isomorphism between six-dimensional Lie algebras. Its image is open, and is closed by compactness of $S^3 \times S^3$. Since $\mathrm{SO}(4)$ is connected, the map is surjective and is the asserted two-sheeted covering. The nontrivial deck transformation is translation by $(-1, -1)$, which is homotopic to the identity since $S^3 \times S^3$ is connected. Formula (2.6) says that both transfer composites are multiplication by two. Therefore the covering induces a homology isomorphism over $\mathbb{F}_5$. Applying ((ii)) to $S^3 \times S^3$ gives $H_5(\mathrm{SO}(4); \mathbb{F}_5) = 0$. This six-manifold has no homology above degree six [3, Theorem 3.26].

The last-column map $\mathrm{SO}(5) \to S^4$ is a locally trivial bundle with fiber $\mathrm{SO}(4)$ [3, Example 4.55]. Cover S^4 by two slightly enlarged open hemispheres U_+ and U_-. The bundle is trivial over each: stereographic coordinates from the opposite pole give a tangent frame; Gram–Schmidt orthonormalization and an orientation choice, followed by adjoining the unit normal, give a smooth section of the frame bundle. Their intersection retracts to S^3, and a trivialization over

U_+ restricts to one over that intersection. Consequently the inverse images have homotopy type $\mathrm{SO}(4)$ and their intersection has homotopy type $\mathrm{SO}(4)\times S^3$. The relevant Mayer–Vietoris terms from ((iv)) are

$$0 = H_9(\mathrm{SO}(4);\mathbb{F}_5)^{\oplus 2} \longrightarrow H_9(\mathrm{SO}(5);\mathbb{F}_5) \longrightarrow H_8(\mathrm{SO}(4)\times S^3;\mathbb{F}_5) = 0,$$

where ((iii)) gives

$$H_8(\mathrm{SO}(4)\times S^3;\mathbb{F}_5) = H_8(\mathrm{SO}(4);\mathbb{F}_5)\oplus H_5(\mathrm{SO}(4);\mathbb{F}_5) = 0.$$

This proves the first assertion.

For the second, let E_0 be the union of fibers over the vertices. Pull the bundle back to each characteristic interval of an edge and trivialize it there. This description also applies to loop edges: their two endpoint fibers are attached to the appropriate vertex fibers by possibly different identifications. Short open stars of the vertices lift to a neighborhood of E_0 which deformation retracts onto E_0: in each incident edge chart, contract the interval coordinate towards its endpoint. These contractions fix the endpoint fibers and hence agree with their gluing maps. Thus (E, E_0) is a good pair, as required in Lemma 2.5. After collapsing E_0, the quotient is a wedge, one summand per edge, of the spaces

$$(\mathrm{SO}(5)\times[0,1])/(\mathrm{SO}(5)\times\{0,1\}).$$

The image of the two endpoint collars in each summand is a neighborhood that contracts to its basepoint, so the wedge also satisfies the good-pair hypothesis. Formulas ((i)) and ((i)), followed by ((iii)), give

$$\begin{aligned} H_{10}(E, E_0;\mathbb{F}_5) &\cong \bigoplus_{\text{edges}} H_{10}(\mathrm{SO}(5)\times[0,1], \mathrm{SO}(5)\times\{0,1\};\mathbb{F}_5) \\ &\cong \bigoplus_{\text{edges}} H_9(\mathrm{SO}(5);\mathbb{F}_5) = 0. \end{aligned}$$

The exact sequence ((ii)) shows that $H_{10}(E_0;\mathbb{F}_5)\to H_{10}(E;\mathbb{F}_5)$ is onto. This argument allows nontrivial monodromy and does not require the bundle over T to be trivial. □

Restrict the representation ρ to C_5. A five-cycle is an even permutation, so $\rho(C_5)\subset\mathrm{SO}(U)$. We use the actions

$$A \longmapsto A\rho(\sigma)^{-1}, \qquad L \longmapsto \rho(\sigma)L$$

on matrices and relation lines. Set $M = \mathrm{SO}(5)/C_5$. Right translations preserve orientation, so M is a connected closed oriented ten-manifold and $[M]\neq 0$ in $H_{10}(M;\mathbb{F}_5)$.

Lemma 4.2. *For $0<\delta<1/6$, let*

$$E_\delta = (\mathrm{SO}(5)\times K_\delta)/C_5, \qquad p_\delta : E_\delta \to M, \quad [V, L] \longmapsto [V],$$

where C_5 acts diagonally by $(V, L)\mapsto (V\rho(\sigma)^{-1}, \rho(\sigma)L)$. Then

$$(p_\delta)_* : H_{10}(E_\delta;\mathbb{F}_5) \longrightarrow H_{10}(M;\mathbb{F}_5) \quad \textit{is zero.}$$

Proof. The equivariant maps and homotopies in Lemma 3.7, applied only to the second factor, give a homotopy equivalence over M between E_δ and

$$E_\Gamma = (\mathrm{SO}(5) \times \bar{\Gamma})/C_5.$$

The latter projects to $\bar{\Gamma}/C_5$ as a bundle with fiber $\mathrm{SO}(5)$, because the action on $\bar{\Gamma}$ is free. To see the local trivialization explicitly, take an evenly covered neighborhood J in $\bar{\Gamma}/C_5$ and a lift $s : J \to \bar{\Gamma}$. Every orbit over J has a unique representative $(V, s(x))$, giving $\mathrm{SO}(5) \times J$ as a product chart. Different lifts change V by a right translation which is locally constant on the base (constant on each connected overlap). Thus this is a bundle even if the covering of the graph is nontrivial. Lemma 4.1 says that its tenth homology is generated by vertex-fiber classes. On each such fiber, the map to M is the five-sheeted covering $\mathrm{SO}(5) \to \mathrm{SO}(5)/C_5$. Its degree is five, which is zero in $\mathbb{F}_5$. The homotopy equivalence from E_δ to E_Γ preserves the first-coordinate projection to M throughout both homotopies, so this vanishing is precisely the asserted vanishing of $(p_\delta)_*$. □

4.2. **The determinant cut.**

We use the preceding homological vanishing to rule out a bordism with the rank and relation constraints established in Section 3. Applying this obstruction to the configuration bordism proves Theorem 1.1.

Lemma 4.3. *There is no compact oriented smooth eleven-manifold B with the following data and properties: a free smooth action of C_5 preserving orientation, and a smooth equivariant map $A : B \to \mathrm{Hom}(U, \mathbb{R}^5)$ satisfying*

(i) *the entire boundary is equivariantly identified with $O(U, \mathbb{R}^5)$, and A restricts there to the inclusion;*
(ii) $\operatorname{rank} A(b) \geq 4$ *for all $b \in B$;*
(iii) $\ker A(b) \in K_{\mathrm{bal}}$ *whenever* $\operatorname{rank} A(b) = 4$.

Proof. We divide the proof into three steps.

Step 1. Define the polar map and prove rank loss.

Suppose such B and A exist, and set $d(b) = \det A(b)$. This is C_5-invariant and equals $+1$ and -1 on the two boundary components. For a positive-determinant matrix define its positive polar factor

$$V(A) = A(A^*A)^{-1/2} \in \mathrm{SO}(5). \tag{4.1}$$

It is smooth. For $R = \rho(\sigma)$, we have

$$(AR^{-1})^*(AR^{-1}) = R(A^*A)R^{-1}, \qquad V(AR^{-1}) = V(A)R^{-1}.$$

Indeed, the inverse positive square root of the first expression is $R(A^*A)^{-1/2}R^{-1}$, which gives the second identity and proves equivariance. We use V only where d is strictly positive.

If d never vanishes, the part $\{d > 0\}$ is compact and has boundary $\mathrm{SO}(5)$. Its quotient maps to M by (4.2), restricting to the identity on the boundary. Formula (2.6) would give $[M] = 0$, a contradiction. Hence $D_0 = d^{-1}(0)$ is nonempty.

Step 2. Extend the kernel line with a uniform balance margin.

Order the singular values as $s_1 \geq \cdots \geq s_5$. Compactness and the rank bound give $s_4 \geq a > 0$ on B. Since $|d| = s_1 s_2 s_3 s_4 s_5$, on $|d| \leq t$ we have $s_5 \leq t/a^4$. Choose $t < a^5/2$. Then $s_5 < a/2$ while $s_4 \geq a$, so the least eigenvalue of A^*A is simple and separated by a uniform gap from the others. Its eigenspace $L(b)$ is a continuous line on this neighborhood of D_0. One way to check continuity is to use its spectral projection

$$\Pi(b) = \frac{1}{2\pi \mathrm{i}} \int_{|z|=a^2/2} \big(zI - A(b)^* A(b)\big)^{-1} \, dz.$$

The contour is traversed counterclockwise and the resolvent is taken on the complexification of U. Diagonalizing the real symmetric matrix shows that the contour encloses only the least eigenvalue and that Π is its real orthogonal projection. It has rank one and depends smoothly on b on the open spectral-gap region. Conjugating A^*A by $\rho(\sigma)$ conjugates Π by the same matrix, so L is equivariant. On D_0, it is exactly $\ker A$.

For a line $L = \mathbb{R}c \subset U$ put

$$\mu(L) = 1 - \frac{\mathrm{osc}(c)}{S(c)}.$$

This is continuous and independent of the nonzero representative. It is positive on $L(D_0)$ by assumption. Let $m_0 > 0$ be its minimum there. Shrink t so that $\mu(L(b)) \geq m_0/2$ whenever $0 \leq d(b) \leq t$, and choose $0 < \delta < \min(1/6, m_0/2)$. Then

$$L(b) \in K_\delta, \qquad 0 \leq d(b) \leq t. \tag{4.2}$$

Indeed, failure for every smaller t would give a sequence b_k with $d(b_k) \downarrow 0$ and $\mu(L(b_k)) < m_0/2$. A subsequence converges in B to a point of D_0, where continuity would contradict the defining minimum m_0.

Step 3. Cut at a regular level and compare boundary classes.

Choose a regular value $0 < \varepsilon < \min(t, 1)$ of d and set

$$B_\varepsilon = \{b \in B : d(b) \geq \varepsilon\}, \qquad Z_\varepsilon = d^{-1}(\varepsilon), \qquad Y_\varepsilon = B_\varepsilon / C_5.$$

Apply Lemma 2.2 to d on the interior of B to obtain such a value. Since $d = \pm 1$ on the original boundary, the level is disjoint from that boundary, so Y_ε is a compact oriented manifold with boundary $M \sqcup (Z_\varepsilon/C_5)$. There are no corners: near the original boundary, $d = \pm 1$ on that boundary and continuity keeps the new level away from it; at the new level, regularity gives an ordinary half-space chart. Lemma 2.7 gives the quotient charts, orientations and boundary identification. Polar decomposition induces $P : Y_\varepsilon \to M$ which is the identity on M. On the other boundary its restriction factors as

$$Z_\varepsilon/C_5 \longrightarrow E_\delta \xrightarrow{p_\delta} M,$$

$$[b] \longmapsto \Big[V\big(A(b)\big), L(b)\Big] \longmapsto \Big[V\big(A(b)\big)\Big].$$

This is well defined by (4.2) and equivariance. Lemma 4.2 makes its map on tenth homology zero. The level may be empty or disconnected; an empty level contributes zero, and a nonempty compact level has finitely many components. This holds for every component of Z_ε/C_5, with its induced orientation, and therefore for their sum. On the connected Euclidean component, the induced orientation is either the standard orientation of M or its negative, so the identity map contributes exactly $\pm[M]$. Applying (2.6) to P gives

$$0 = P_*[\partial Y_\varepsilon] = \pm[M] + 0,$$

contrary to $[M] \neq 0$. □

Proof of Theorem 1.1. Suppose a norm on $\mathbb{R}^5$ has no six-point equilateral set. By Lemma 3.2, choose a smooth strongly convex norm N^1 with the same property. Let N^0 be Euclidean. Both endpoints are regular, the second because its fiber is empty. Lemma 3.6 gives a compact oriented eleven-manifold $\mathcal{B}$ whose entire boundary is $O(U, \mathbb{R}^5)$. The C_5-action is free and orientation preserving.

Assign to each $(t, y) \in \mathcal{B}$ the matrix A_y from (3.2). This map is smooth, equals the boundary isometry at the Euclidean endpoint, and satisfies

$$A_{\sigma \cdot y} = A_y \rho(\sigma)^{-1}.$$

All norms on the path are strictly convex. Lemma 3.5 therefore gives $\operatorname{rank} A_y \geq 4$, and every rank-four kernel lies in K_{bal}. This contradicts Lemma 4.3. Thus the six points exist. Translating one of them to zero and dividing by their common distance gives the stated normalization. □

Remark 4.4. For $n \geq 6$, the rank bound and graph reduction above give no corresponding control of $(n+1)$-point configurations. With seven labels, $c = (1/3, 1/3, 1/3, -1/2, -1/2, 0, 0)$ satisfies $S(c) = 1$ and $\operatorname{osc}(c) = 5/6$, but four sign cells meet at its two zero coordinates. Thus the pairwise-intersection construction of Lemma 3.7 fails; appending zeros gives the same example with more labels. Without suitable rank and relation-space control, Lemma 4.1 does not reduce the determinant-level homology to fiber classes, so the covering-degree cancellation is unavailable. This limitation does not decide the conjecture in the remaining dimensions.

References

[1] L. Danzer, B. Grünbaum, Über zwei Probleme bezüglich konvexer Körper von P. Erdős und von V. L. Klee, Math. Z. 79 (1962) 95–99. 1

[2] V. Guillemin, A. Pollack, Differential topology, Prentice-Hall, Englewood Cliffs, NJ, 1974. 3

[3] A. Hatcher, Algebraic topology, Cambridge University Press, Cambridge, 2002. 4, 5, 6, 15

[4] T. Kobos, An alternative proof of Petty's theorem on equilateral sets, Ann. Polon. Math. 109(2) (2013) 165–175. 1

[5] J. M. Lee, Introduction to smooth manifolds, 2nd ed., Graduate Texts in Mathematics, vol. 218, Springer, New York, 2013. See also the author's corrections, entries dated 28 September 2019 for p. 146. 3, 6, 11

[6] V. V. Makeev, Equilateral simplices in normed 4-space, Zap. Nauchn. Sem. POMI 329 (2005) 88–91; English translation: J. Math. Sci. 140(4) (2007) 548–550. 1

[7] C. M. Petty, Equilateral sets in Minkowski spaces, Proc. Amer. Math. Soc. 29(2) (1971) 369–374. 1, 2

[8] A. Schürmann, K. J. Swanepoel, Three-dimensional antipodal and norm-equilateral sets, Pacific J. Math. 228(2) (2006) 349–370. 2, 10

[9] K. J. Swanepoel, Equilateral sets in finite-dimensional normed spaces, in: D. Girela Álvarez, G. López Acedo, R. Villa Caro (eds.), Seminar of Mathematical Analysis, Colecc. Abierta, vol. 71, Universidad de Sevilla, Seville, 2004, pp. 195–237. 1

[10] K. J. Swanepoel, R. Villa, Maximal equilateral sets, Discrete Comput. Geom. 50(2) (2013) 354–373. 2

[11] A. C. Thompson, Minkowski geometry, Encyclopedia of Mathematics and its Applications, vol. 63, Cambridge University Press, Cambridge, 1996. 1, 7

Appendix A. **Prompt provided to hyprover**

The following prompt was written by the author and provided to hyprover.

Five-dimensional Petty problem

Objective

Prove that every five-dimensional real normed vector space contains six equilateral points. Precisely, for every norm $\|\cdot\|$ on $\mathbb{R}^5$, prove that there exist distinct points $x_0, \ldots, x_5$ such that

$$\|x_i - x_j\| = 1 \qquad (0 \le i < j \le 5).$$

The norm is arbitrary; affine independence of the six points is not required.

Research hint

Seek an appropriate differential-topological reformulation while preserving the original geometric constraints. Differential geometry and algebraic topology may help; consult Lu–Yang's *A Solution to Banach's Isometric Conjecture* for methodological inspiration, not a ready-made argument.

Working requirements

- Work in English and preserve the exact target throughout.
- Supply a complete, independently checkable proof, including all essential intermediate arguments and precise references for external results.
- Distinguish proved statements, conjectures, failed approaches, and unresolved gaps. Never replace a missing argument with a claim of verification.
- Do not count special cases, conditional reductions, numerical evidence, or a proof sketch as a solution.